



React



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Agenda

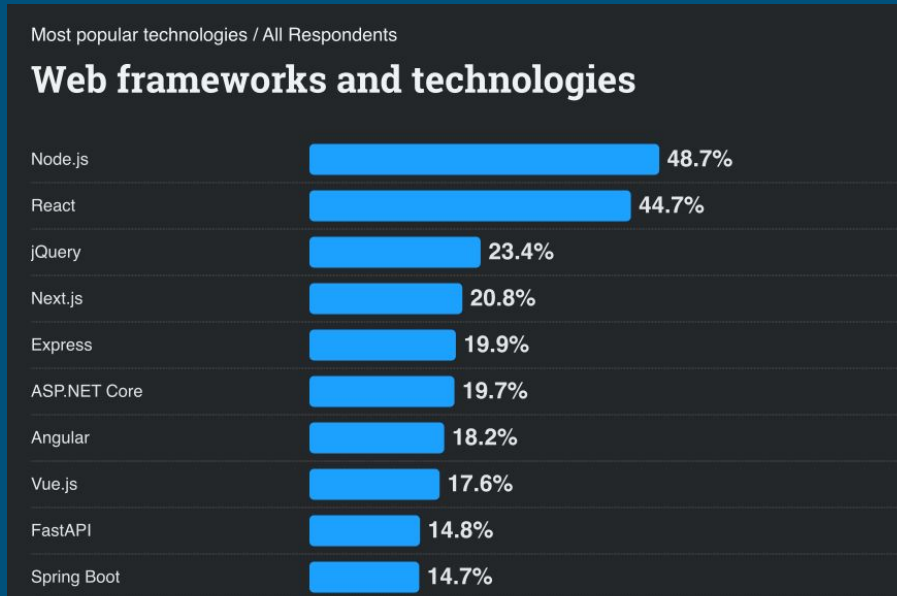
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Intro

- React is a JavaScript library for building UIs
- UIs are composed by creating isolated components
- Provides APIs for allowing your UI to react to changes in internal state
- Return markup from your JavaScript functions with JSX

History

- Created by Jordan Walke at Facebook
- Used internally for Facebook and Instagram
- Open sourced in May 2013
- One of the most popular web libraries
- Introduced React Hooks in 2019



Demo

```
src > components >  MyComponent.tsx > ...  
1   export default function MyComponent() {  
2     const name = 'Ryan';  
3  
4     return <p>My name is {name}</p>;  
5   }
```

```
src >  App.tsx > ...  
1   import MyComponent from './components/MyComponent';  
2  
3   export default function App() {  
4     return <MyComponent />;  
5   }
```

My name is Ryan

Demo

src > components >  MyComponent.tsx > ...

```
1   export default function MyComponent({ name }: { name: string }) {  
2     |   return <p>My name is {name}</p>;  
3   }
```

src >  App.tsx > ...

```
1   import MyComponent from './components/MyComponent';  
2  
3   export default function App() {  
4     |   return <MyComponent name="Ryan" />;  
5   }
```

Demo

src > components >  StateExample.tsx > ...

```
1  import { useState } from 'react';
2
3  export default function StateExample() {
4    const [count, setCount] = useState(0);
5
6    function incrementCount() {
7      setCount((prev) => prev + 1);
8    }
9
10   return <button onClick={incrementCount}>Count: {count}</button>;
11 }
```

Count: 0

Count: 1

Count: 2

Demo

src > components >  EffectExample.tsx > ...

```
1  import { useEffect, useState } from 'react';
2
3  export default function EffectExample() {
4    const [count, setCount] = useState(0);
5
6    useEffect(() => {
7      console.log(`count is now ${count}`);
8    }, [count]);
9
10   function incrementCount() {
11     setCount((prev) => prev + 1);
12   }
13
14   return <button onClick={incrementCount}>Count: {count}</button>;
15 }
```

count is now 0

count is now 1

count is now 2

count is now 3

count is now 4

count is now 5

count is now 6

Demo

src > components >  EffectExample2.tsx > ...

```
1  import { useEffect, useState } from 'react';
2
3  export default function EffectExample2() {
4    const [count, setCount] = useState(0);
5
6    useEffect(() => {
7      const id = setInterval(() => {
8        setCount((prev) => prev + 1);
9      }, 1000);
10
11      return () => clearInterval(id);
12    }, []);
13
14    return <p>{count}</p>;
15  }
```

2

5

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Demo

src > components >  ServerComponentExample.tsx > ...

```
1  import db from './database';
2
3  export default async function ServerComponentExample({ userId }: { userId: string }) {
4    const { name } = await db.users.get(userId);
5
6    return <p>Username: {name}</p>;
7  }
```

Potential Usage in Capstone Projects

- Your app probably has a frontend
- Your frontend is probably on the web
- React Native allows iOS/Android apps to be written with React
 - Has potential to share codebase between web and mobile
- Widely used, your team members likely have experience with React

Summary

- React is a library for creating reactive UIs for your web apps
- UIs are composed of components
- Parameters are passed between components via props
- React Hooks provide components with interactivity
- React Server Components brings React to your backend

References

- <https://react.dev>
- [https://en.wikipedia.org/wiki/React_\(software\)](https://en.wikipedia.org/wiki/React_(software))